

PROJECT REPORT



TITLE:

SOHO Network Design & Implementation:
Small Office / Home Office Simulation

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Abstract:

This project uses Cisco Packet Tracer to mimic a small office/home office (SOHO) network. VLAN-based departmental segmentation, dynamic IP assignment via DHCP, router-on-a-stick inter-VLAN routing, and wireless connectivity via access point or access points are all implemented in this system. The network supports both wired and wireless hosts and links several departments (such as Admin/IT, Finance/HR, and Customer Service/Reception), each of which is conceptually segregated but capable of communicating when necessary. Before expanding to bigger enterprise/campus networks, the design helps develop fundamental skills by demonstrating fundamental networking concepts (switching, routing, VLANs, DHCP, and wireless) appropriate for small company settings.

Overview and Goals:

1.1 Background:

A straightforward yet effective network that serves several departments or user groups with isolation for security and management is frequently required by small firms, branch offices, or home offices. With few hardware resources, few users, and the requirement for both wired and wireless connectivity, a SOHO-style design presents a realistic scenario. Without the need for physical hardware, students can study networking principles like VLANs, inter-VLAN routing, DHCP, and wireless settings by simulating such networks with Packet Tracer.

1.2 Project Goals:

To meet departmental segmentation, create a small office network infrastructure with just one router, one switch, and one wireless access point.

To isolate traffic and safely manage resources, create distinct VLANs for various departments (such as Admin/IT, Finance/HR, and Customer Service).

Set up IP addressing and subnetting to assign unique subnets to each VLAN, allowing broadcast domains to be clearly separated.

Use router-on-a-stick (sub-interfaces on a router's single physical interface) to implement inter-VLAN routing so that all departments can communicate as needed.

To make management easier, especially for visitors and wifi users, set up DHCP to dynamically assign IP addresses to end-devices (PCs, laptops).

Incorporate wireless connectivity (via an access point) for departments that require WiFi.

To make sure the network operates properly, test and confirm full connectivity between devices across VLANs, including wired and wireless.

Architecture & Design of Networks:

2.1 Overview of Topology and Hierarchical Design:

The network utilizes very little hardware because of the small-office scope:

- One Cisco router that links all VLANs and sub-interfaces
- One Layer-2 switch serves as the access switch that links the access point and wired devices.
- One wireless access point (to connect to WiFi).
- End devices include laptops, PCs, and possibly peripherals like printers.
- This simple arrangement is representative of a typical SOHO setting where affordability and ease of use are important.

2.2 VLAN and Department Segmentation:

Because there are three departments, the network is divided into three VLANs. For example (assuming a base network 192.168.1.0/24 — as in the project scenario) One possible subnetting approach:

- Use subnet mask 255.255.255.192 (i.e., /26) so each subnet supports up to 62 hosts. This gives 4 subnets from 192.168.1.0/24, of which 3 are used for departments, leaving one subnet free (future use / guest / expansion).
- Example mapping:

VLAN / Department	Subnet	Default Gateway (Router sub-interface)
VLAN 10 – Admin/IT	192.168.1.0/26	192.168.1.1
VLAN 20 – Finance/HR	192.168.1.64/26	192.168.1.65
VLAN 30 – Customer Service / Reception	192.168.1.128/26	192.168.1.129

- DHCP pools configured for each subnet so that end devices automatically get IP, gateway, DNS etc.

The above design ensures efficient use of IP space, clear separation of department networks, and ease of management

2.3 Configuring VLANs and Inter-VLAN Routing:

Steps for configuration (on devices or in Packet Tracer):

VLANs (such as VLAN 10, 20, and 30) should be defined on the switch, and access ports should be assigned to VLANs that correspond to department hosts.

To allow multiple VLAN traffic to pass across a single link, configure the switch-to-router link as a trunk port (802.1Q).

Create sub-interfaces on the router's physical interface that is connected to the switch, assign each sub-interface to a VLAN (dot1Q encapsulation), and provide the IP

address that corresponds to the subnet (as default gateway).

In order to facilitate inter-departmental communication while preserving logical separation, enable routing so that packets between VLANs are routed via router.

2.4 DHCP And Wireless Configuration:

This router-on-a-stick technique is effective for small networks because it uses a single physical router interface to control several VLANs..

Set up pools for every VLAN subnet on the router (or a different DHCP server). Network, default router, DNS server (if internal), subnet mask, domain (if required), etc., should all be specified for each pool. Set up a wireless access point (WAP) or port on the switch. In order to deploy wireless clients in the right VLAN (or a dedicated guest VLAN/subnet, depending on requirements), the WAP should be connected via trunk or appropriate VLAN tagging. Set up the WAP's SSID and security (such as WPA2-PSK); make sure wireless clients may connect with wired clients as necessary and receive DHCP addresses.

2.5: Physical vs. Logical/Cabling Layout:

Connect routers, switches, PCs, and WAP via straight-through connections (or automatically in Packet Tracer). Set up switch ports: trunk port for router links, access port for PCs (assigned to VLANs). Packet Tracer's logical form makes network design easy to see; if necessary, you may create a "physical" view (floor-plan style) by arranging devices in various rooms or buildings. This is helpful for documentation and presentations. It is easier to simulate realistic deployment when logical and physical views are used.

Cisco Packet Tracer implementation:

Step-by-step description based on typical SOHO network implementation (as in project):

3.1 Initial Setup:

- Open Cisco Packet Tracer.
- Add required devices: 1 router, 1 switch, 1 wireless access point, necessary number of PCs/laptops.

3.2 Switch Configuration:

- Create VLANs (e.g. VLAN 10, 20, 30).
- Assign switch ports to VLANs (access ports) for PCs.
- Configure the port connecting to router as trunk (802.1Q).

3.3 Router Configuration (Router-on-a-Stick):

- On router, create sub-interfaces under the main interface (e.g. GigabitEthernet0/0.10, 0/0.20, 0/0.30).

- For each sub-interface, set encapsulation dot1Q <VLAN_ID> and assign IP address (default gateway for that VLAN).
- Enable sub-interfaces (no shutdown).

3.4 DHCP Configuration:

- On router (or external DHCP server), create DHCP pools — one per VLAN subnet.
- Define parameters: network address, subnet mask, default gateway, DNS server (if needed), domain name (optional).
- Ensure DHCP is enabled, and save configuration.

3.5 Wireless Access Setup:

- Connect WAP to switch using an appropriate port (trunk or access, depending on design).
- Configure WAP: define SSID, security mode (WPA2-PSK), password, possibly VLAN tagging (if wireless clients belong to a specific VLAN).
- Enable wireless; ensure devices can connect, get DHCP IP, and access network.

3.6 Host Configuration / Testing:

- On PCs / laptops: set to obtain IP automatically (DHCP) — verify they receive correct IP, subnet mask, gateway.
- Test connectivity: ping between devices in same VLAN; ping across VLANs; test internet access (if simulated), test wireless connectivity.
- Document network: router/sub-interface config, switch VLAN config, DHCP config, wireless config.

Findings and Discussion:

4.1 Benefits and Strengths:

Simplicity and Cost-Effectiveness: Small offices and branches can benefit from the low cost of minimal gear (one router, one switch).

Logical Segmentation: VLANs offer enhanced network organization, security, and logical separation.

Dynamic IP Management: DHCP makes it easier to configure and manage devices, especially when users and devices come and go on a frequent basis.

Wireless Flexibility: Mobility and convenience are made possible via wireless access (laptops, guest devices).

Learning/Training Value: This configuration illustrates basic networking principles for students or small businesses, which can serve as a foundation for further education and growth.

4.2 Restrictions and Warnings:

Scalability: Small networks can benefit from this approach. One router and one switch may become bottlenecks when a company expands (many devices, numerous VLANs).

Redundancy and Fault Tolerance: Without redundancy, a single point of failure (a switch or router) might bring down the entire network.

Security Restrictions: Simple setup; no sophisticated security capabilities (firewall, intrusion detection, access control lists, segmentation beyond VLANs).

Wireless Coverage & Capacity: One WAP may not be able to cover every area in larger offices; each WAP can only handle a certain number of customers; failover and load-balancing are not available.

Limitations on Resources: Real hardware is required for production, and performance may deteriorate with heavy traffic (multiple devices, huge data loads). Additionally, Packet Tracer's simulation does not accurately represent real-world problems like latency, interference, hardware constraints, and simultaneous load.

4.3 Possible Enhancements and Expansions:

When expanding or improving the design, take into account:

Routing is made easier and performance is enhanced by using a multi-layer switch (L3 switch) rather than a router and switch.

Redundancy can be added by using redundant links, a supplementary router or switch, the spanning-tree protocol (for loops), link aggregation, and port channels.

Access control lists (ACLs), VLAN ACLs, external or internal firewalls, network segmentation outside of VLANs, and VPNs for distant users are all examples of security implementation.

Multiple WAPs throughout the office, a real or simulated wireless controller, guest VLAN, SSID segmentation, bandwidth management, and QoS are all examples of expanding wireless.

Internal servers (file server, print server, DNS, DNS, DHCP, possibly VPN server), monitoring, logging, and backup network are examples of extra services that can be added.

Documentation and version control: maintain configuration scripts, address plans, VLAN/sub-interface assignment documentation, network diagrams (logical and physical), security policies, etc.

Results & Verification:

If the aforementioned setting is implemented correctly in Packet Tracer, the following outcomes should be seen:

Each VLAN's devices can ping their default gateway (router sub-interface) and get accurate IP addresses dynamically through DHCP.

Inter-VLAN communication is made possible via router-on-a-stick routing, which allows devices from various VLANs to communicate (ping or access common resources).

Wireless devices that are connected using WAP are able to communicate with wired devices and successfully obtain IP via DHCP.

VLAN separation is preserved by isolating broadcast domains according to VLANs, which lowers broadcast traffic and improves security and performance isolation.

The network is straightforward but effective, making it appropriate for home offices, branch offices, and small businesses.

These findings verify that the design satisfies the following project goals:

departmental isolation + inter-connectivity, DHCP automation, WLAN support, VLAN segmentation, and routing.

Conclusion:

This SOHO-network simulation experiment using Cisco Packet Tracer shows that it is possible to create a working, VLAN-segmented, routed, wireless-capable small office network that is appropriate for home offices, branch offices, or small businesses, even with very little gear. VLAN-based departmental segregation, dynamic IP management, inter-VLAN routing, and wireless connectivity are all successfully achieved by the project.

For students and network learners to grasp fundamental network architecture concepts (switching, routing, VLANs, DHCP, wireless), it provides a solid foundational exercise. Additional improvements (redundancy, security, and speed optimization) would be necessary for real-world deployment or scaling.

References:

- SOHO Network Design & Implementation Project (Project #2) — design requirements and topology description.
- SOHO Networking Project – example implementation and subnetting/VLAN/DHCP/wireless configuration details
- Documentation on network switch roles and why switches are used in LANs.
- Overview & role of Cisco Packet Tracer as a simulation tool for network design and testing